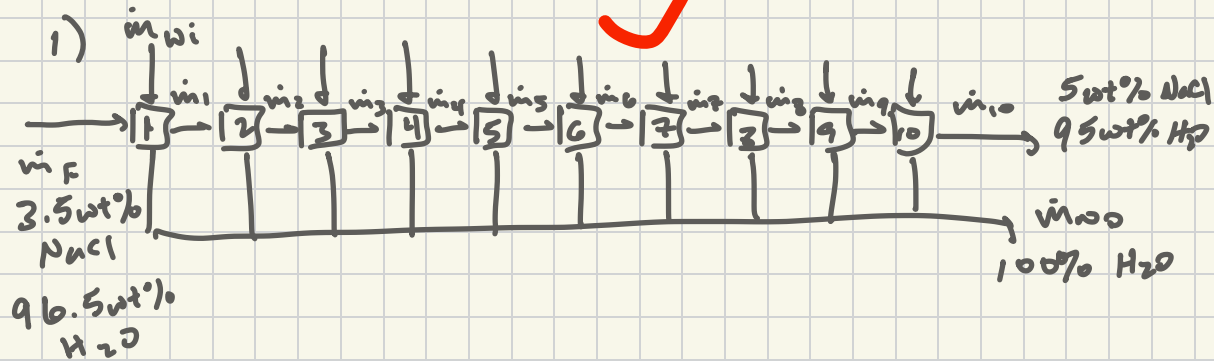


# Homework 4



$$\text{Total: } m_F + 10 m_{wi} = m_{10} + m_{w0}$$

$$H_2O: 0.965 m_F + 10 m_{wi} = 0.95 m_{10} + m_{w0}$$

$$NaCl: 0.035 m_F = 0.05 m_{10}$$

Basis:  $m_F = 1 \text{ kg/s}$

$$0.035 = 0.05 m_{10} \quad m_{10} = 0.7$$

$$1 + 10 m_{wi} = 0.7 + m_{w0}$$

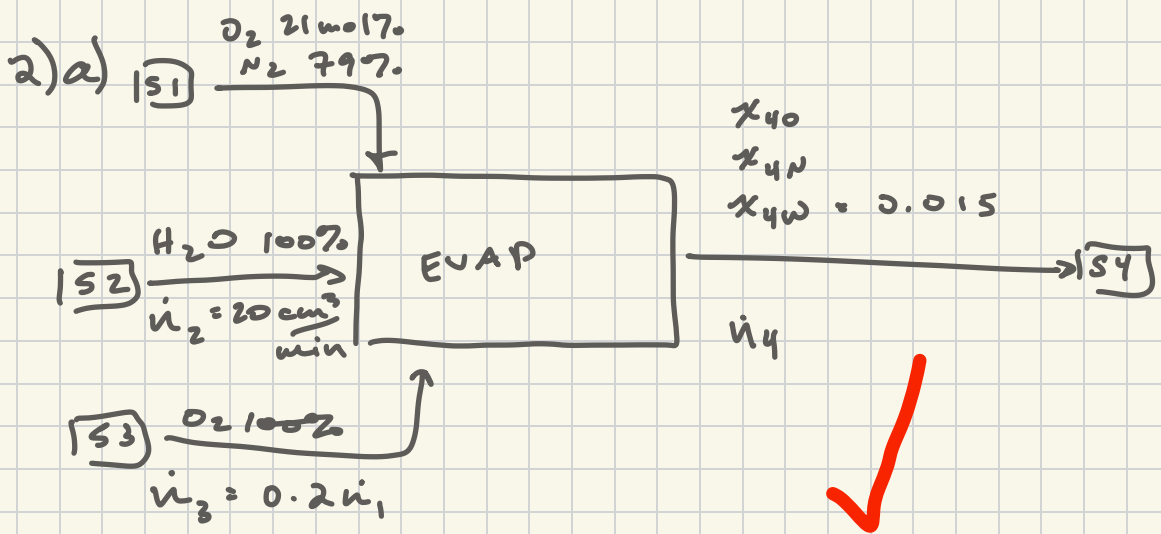
$$0.965 + 10 m_{wi} = 0.95(0.7) + m_{w0}$$

$$m_{w0} = 10 m_{wi} + 0.3$$

$$0.3$$

Didn't finish c, no attempt at d. -8 pts

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b) DoF:

# of stream variables: 7  $\rightarrow$  7

# spec. flow rates: 2

# spec. stream comp: 2

# bal. equations: 3

} 7

$7 - 7 = 0 \therefore$  well-posed

c)  $M_w = 18.015 \text{ g/mol}$   $\rho = 1.00 \text{ g/cm}^3$

$$i_1 + i_2 + i_3 = i_4$$

$$\text{H}_2\text{O}: i_2 = 0.015 i_4 = 20$$

$$* i_3 = 0.2 i_1$$

$$\text{O}_2: 0.21 i_1 + i_3 = x_{4\text{O}} i_4$$

$$i_2 = 20 \text{ cm}^3/\text{min}$$

$$\text{N}_2: 0.79 i_1 = x_{4\text{N}} i_4$$

$$20 = 0.015 \dot{v}_4 \quad \dot{v}_4 = 1333.33$$

$$1.2 \dot{v}_1 + 20 = 1333.33$$

$$\dot{v}_1 = 1094.44 \quad \text{X}$$

$$\dot{v}_3 = 218.88 \quad \text{X}$$

$$0.21(1094.44) + 218.88 = x_{4O}(1333.33)$$

$$x_{4O} = 0.3365 \quad \checkmark$$

$$0.79(1094.44) = x_{4N}(1333.33)$$

$$x_{4N} = 0.6485 \quad \checkmark$$

So...

$$\dot{v}_1 = 1094.44 \text{ cm}^3/\text{min}$$

$$\dot{v}_2 = 20 \text{ cm}^3/\text{min}$$

$$\dot{v}_3 = 218.88 \text{ cm}^3/\text{min}$$

$$\dot{v}_4 = 1333.33 \text{ cm}^3/\text{min}$$

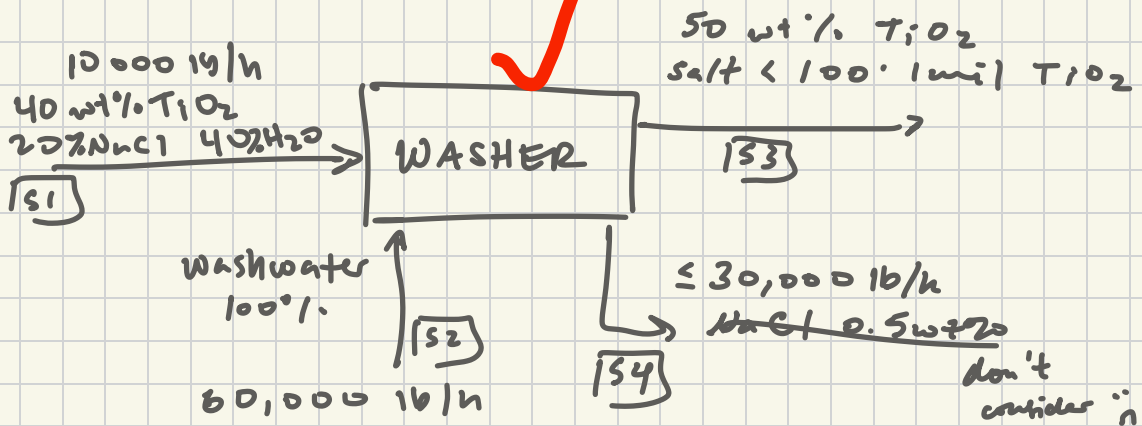
$$x_{4O} = 0.3365$$

$$x_{4N} = 0.6485$$

These are volumetric flow rates but the compositions are in terms of mol %. Needed to convert to molar flow using density of water and molar mass, then solve. -3 pts

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### 3) Murphy 2.66



DoF:

$$\begin{array}{lcl} 4a: 9 & & 5a: 2 \\ 4b: 0 & \left. \vphantom{\begin{array}{l} 4a: 9 \\ 4b: 0 \end{array}} \right\} 9 & 5c: 0 \\ & & 5d: 3 \end{array} \left. \vphantom{\begin{array}{l} 5a: 2 \\ 5c: 0 \\ 5d: 3 \end{array}} \right\} 9$$

$$9 - 9 = 0 \therefore \text{well posed}$$

balances:

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3 + \dot{m}_4$$

$$\text{TiO}_2: 0.4 \dot{m}_1 = 0.5 \dot{m}_3$$

$$10\,000 (0.4) = 0.5 \dot{m}_3$$

$$\dot{m}_3 = 8000 \text{ lb/hr}$$

$$\dot{m}_4 = 10\,000 + 60\,000 - 8000$$

$$\dot{m}_4 = 62\,000 \text{ lb/hr}$$

No wt%? -1 pt

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#### 4) Murphy 2.69

DoF:

# indep. stream variables: 25  
# indep. system variables: 2 } 27

# specified flows: 1  
# specified comp.: 4  
# performance spec: 2  
# mat. bal. equat.: 20 } 27

$$27 - 27 = 0$$

∴ properly specified

1. Mixer: ( $\dot{n}_1 + \dot{n}_3 + \dot{n}_{10} = \dot{n}_2$ )

$$\hookrightarrow x_{p1} \dot{n}_1 + x_{p10} \dot{n}_{10} = x_{p2} \dot{n}_2$$

$$x_{i1} \dot{n}_1 + x_{i10} \dot{n}_{10} = x_{i2} \dot{n}_2$$

$$x_{b1} \dot{n}_1 + \dot{n}_3 + x_{b10} \dot{n}_{10} = x_{b2} \dot{n}_2$$

2. Reactor: ( $\dot{n}_2 = \dot{n}_3$ )

$$\hookrightarrow x_{p2} \dot{n}_2 = x_{p3} \dot{n}_3 + A + B$$

$$x_{i3} \dot{n}_3 = x_{i2} \dot{n}_2$$

$$x_{b3} \dot{n}_3 = x_{b2} \dot{n}_2 - A$$

$$x_{c3} \dot{n}_3 = A - B$$

$$x_{d3} \dot{n}_3 = B$$

3. V/L Separator: ( $\dot{n}_3 = \dot{n}_q + \dot{n}_4$ )

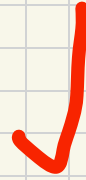
$$x_{p3} \dot{n}_3 = x_{p9} \dot{n}_q$$

$$x_{i3} \dot{n}_3 = x_{i9} \dot{n}_q$$

$$x_{b3} \dot{n}_3 = x_{b4} \dot{n}_4$$

$$x_{c3} \dot{n}_3 = x_{c4} \dot{n}_4$$

$$x_{d3} \dot{n}_3 = x_{d4} \dot{n}_4$$



4. Splitter: ( $\dot{n}_q = \dot{n}_{10} + \dot{n}_{11}$ )

$$x_{p9} \dot{n}_q = x_{p10} \dot{n}_{10} + x_{p11} \dot{n}_{11}$$

$$x_{i9} \dot{n}_q = x_{i10} \dot{n}_{10} + x_{i11} \dot{n}_{11}$$



5. Distillation col. 1: ( $\dot{n}_4 = \dot{n}_3 + \dot{n}_5$ )

$$x_{b4} \dot{n}_4 = x_{b3} \dot{n}_3$$

$$x_{c4} \dot{n}_4 = x_{c5} \dot{n}_5$$

$$x_{d4} \dot{n}_4 = x_{d5} \dot{n}_5$$



6. Distillation col. 2: ( $\dot{n}_5 = \dot{n}_6 + \dot{n}_7$ )

$$x_{c5} \dot{n}_5 = x_{c6} \dot{n}_6$$

$$x_{d5} \dot{n}_5 = x_{d7} \dot{n}_7$$



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